
Curriculum Vitae

Felix Aron Dannert

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Education

2021–2025	<i>Ph.D. in Astrophysics</i> (Eidgenössische Technische Hochschule (ETH) Zürich)
2018–2021	<i>M.Sc. in Physics</i> (ETH Zürich)
2015–2018	<i>B.Sc. in Physics</i> (Universität Konstanz)

Experience

since 2025	<i>Lead Instrument Scientist – LIFE Space Mission at (ETH Zürich) Website</i>
since 2025	<i>President – Akademische Fluggruppe Zürich (AFG) Website</i>
2023–2025	<i>Vice-President – Akademische Fluggruppe Zürich (AFG)</i>
2021	<i>Co-Initiator – NAUTILUS project Website</i>
2019–2020	<i>Board member – Akademische Raumfahrt Initiative Schweiz (ARIS) Website</i>
2019	<i>Systems Engineer – Sounding rocket HEIDI Website</i>

Teaching experience

2024	<i>Teaching assistant – Exoplanets (ETH Zürich)</i>
2021–2023	<i>Teaching assistant – Physics I; 3 semesters (ETH Zürich)</i>
since 2021	<i>Co-Initiator & Co-Organizer – Lecture Series: Space Research and Exploration; 4 semesters (ETH Zürich)</i>
2017	<i>Teaching assistant – Analysis I (Universität Konstanz)</i>

Poster presentations, talks, and workshops

09/2023	<i>Participation</i> – LIFE science workshop, FU Berlin, Germany
06/2023	<i>Invited participation</i> – Breakthrough Discuss, UC Santa Cruz, USA
06/2023	<i>Contributed Talk</i> – 242nd AAS Meeting , Albuquerque, USA (online)
11/2022	<i>Contributed Poster</i> – invited KISS Workshop , Caltech, USA
05/2022	<i>Contributed Talk & Poster</i> – Exoplanets IV , Las Vegas, USA
04/2022	<i>Contributed Talk</i> – Junior Researchers' Assembly (JURA III), Switzerland
09/2021	<i>Contributed Talk & Poster</i> – EPSC 2021 , Online

Supervision

2024	<i>Master's Thesis</i> – Tim Mosimann : Inferring basic circumplanetary disk properties using JWST/MIRI photometry
2024	<i>Semester Thesis</i> – Olivier Jungo : Infrared Stellar Variability on Sub-Hour Timescale in the Context of Nulling Interferometry
2024	<i>Semester Thesis</i> – Laura Ruther : Analysis of noise confusion considering exoplanets observed by the LIFE mission
2023	<i>Master's Thesis</i> – Vania Vincenzi : Multi-planet systems in LIFE
2023	<i>Master's Thesis</i> (Co-supervision) – Rick Röthlisberger : HWO Meets LIFE: Synergies Between Future Exoplanet Direct Imaging Flagship Missions in Reflected Light and Thermal Emission
2023	<i>Master's Thesis</i> (Co-supervision) – Caroline Strähl : The role of technical requirements in signal extraction for nulling interferometry in the context of the LIFE mission
2023	<i>Semester Thesis</i> (Co-supervision) – Mariasole Agazzi : Optimisation LIFE array movements
2023	<i>Semester Thesis</i> (Co-supervision) – Rick Röthlisberger : Targeting the Most Wanted Exoplanets: How to Quantify Requirements for Observation of Specific Exoplanet Types Through LIFE
2023	<i>Semester Thesis</i> – Marcellin Feasson : The LIFE mission: from the search phase to the characterization phase
2022	<i>Semester Thesis</i> – Lukas Felix : A comparison of local zodiacal dust models for future use in the LIFEsim tool
2021	<i>Semester Thesis</i> – Xavier Kervyn : Impact of non-perfect nulls on the detectable population by the LIFE space mission
2021	<i>Semester Thesis</i> – Fabio Holdener : Implementation of Inclined Exozodi Dust Disks Into LIFEsim and Their Impact on LIFE's Capability to Detect Exoplanets

2021	<i>Semester Thesis</i> – Andreas Küttel : Imagining the Second Phase of the LIFE mission
2021	<i>Semester Thesis</i> – Tanja Kaister : Investigating Nearby Known Exoplanets With LIFEsim
2021	<i>Semester Thesis</i> – Christoph Strähl : Effects of time simulation on the signal in LIFEsim
2021	<i>Semester Thesis</i> – Nicole Lehmann : Influence of the baseline on the SNR of detected exoplanets

Scientific community service

2022–2023	<i>Reviewer</i> for NASA Review Panel
2022–2023	<i>Commissioning</i> of VLT/ERIS – Contributed to planning, execution, evaluation and documentation of HCI-mode commissioning
since 2021	<i>Developer</i> and maintainer of open-source package LIFEsim

Awards

2018	<i>VEUK Award</i> – outstanding Bachelor’s degree (Universität Konstanz)
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Awarded telescope time (selection)

2025	<i>Co-investigator</i> – PI: Gabriele Cugno; Circumplanetary disks; MIRI: 60 h
2025	<i>Co-investigator</i> – PI: Valentin Christiaens; Exoplanet direct imaging; SPHERE: 4 h
2024–2025	<i>Co-investigator</i> – PI: Markus Bonse; Exoplanet direct imaging; ERIS: 50 h
2024	<i>Principal investigator</i> – Exoplanet direct imaging; ERIS: 3 h
2024	<i>Co-investigator</i> – PI: Gabriele Cugno; Exoplanet direct imaging; VISIR: 4 h
2022	<i>Co-investigator</i> – PI: Anna Boehle; Exoplanet radial velocity; ESPRESSO: 72 h

Publications

1. **Dannert, F. A.***, M. Ottiger*, S. P. Quanz, R. Laugier, E. Fontanet, A. Gheorghe, O. Absil, C. Dandumont, D. Defrère, C. Gascón, A. M. Glauser, J. Kammerer, T. Lichtenberg, H. Linz, and J. Loicq (2022). “Large Interferometer For Exoplanets (LIFE): II. Signal simulation, signal extraction, and fundamental exoplanet parameters from single-epoch observations.” *A&A*, 664: A22. ISSN: 1432-0746. DOI: [10.1051/0004-6361/202141958](https://doi.org/10.1051/0004-6361/202141958). URL: <http://dx.doi.org/10.1051/0004-6361/202141958>
2. **Dannert, F. A.**, P. A. Huber, T. Birbacher, R. Laugier, M. J. Bonse, E. O. Garvin, A. M. Glauser, V. Oehl, and S. P. Quanz (2025). “Consequences of Non-Gaussian Instrumental Noise in Perturbed Nulling Interferometers.” *ApJ*. DOI: [10.3847/1538-3881/add720](https://doi.org/10.3847/1538-3881/add720)

My work is divided between three major topics: nulling interferometry with LIFE, the high contrast imaging commissioning of the VLT/ERIS instrument and infrared high-contrast imaging science observations with the VLT and JWST.

For my work on LIFE, I contributed to the following peer-review publications:

3. Quanz, S. P., M. Ottiger, E. Fontanet, J. Kammerer, F. Menti, **F. Dannert**, A. Gheorghe, O. Absil, V. S. Airapetian, E. Alei, R. Allart, D. Angerhausen, et al. (2022). “Large Interferometer For Exoplanets (LIFE): I. Improved exoplanet detection yield estimates for a large mid-infrared space-interferometer mission.” *A&A*, 664: A21. ISSN: 1432-0746. DOI: [10.1051/0004-6361/202140366](https://doi.org/10.1051/0004-6361/202140366). URL: <http://dx.doi.org/10.1051/0004-6361/202140366>
4. Kammerer, J., S. P. Quanz, and **F. Dannert** (2022). “Large Interferometer For Exoplanets (LIFE): VI. Detecting rocky exoplanets in the habitable zones of Sun-like stars.” *A&A*, 668: A52. ISSN: 1432-0746. DOI: [10.1051/0004-6361/202243846](https://doi.org/10.1051/0004-6361/202243846). URL: <http://dx.doi.org/10.1051/0004-6361/202243846>
5. Angerhausen, D., M. Ottiger, **F. Dannert**, Y. Miguel, C. Sousa-Silva, J. Kammerer, F. Menti, E. Alei, B. S. Konrad, H. S. Wang, and S. P. Quanz (2023). “Large Interferometer for Exoplanets: VIII. Where Is the Phosphine? Observing Exoplanetary PH₃ with a Space-Based Mid-Infrared Nulling Interferometer.” *Astrobiology*, 23 (2): 183–194. ISSN: 1557-8070. DOI: [10.1089/ast.2022.0010](https://doi.org/10.1089/ast.2022.0010). URL: <http://dx.doi.org/10.1089/ast.2022.0010>

6. Laugier, R., D. Defrère, J. Woillez, B. Courtney-Barrar, **F. A. Dannert**, A. Matter, C. Dandumont, S. Gross, O. Absil, A. Bigioli, G. Garreau, L. Labadie, et al. (2023). “Asgard/NOTT: L-band nulling interferometry at the VLT: I. Simulating the expected high-contrast performance.” *A&A*, 671: A110. ISSN: 1432-0746. DOI: [10.1051/0004-6361/202244351](https://doi.org/10.1051/0004-6361/202244351). URL: <http://dx.doi.org/10.1051/0004-6361/202244351>

7. Carrión-González, Ó., J. Kammerer, D. Angerhausen, **F. Dannert**, A. García Muñoz, S. P. Quanz, O. Absil, C. A. Beichman, J. H. Girard, B. Mennesson, M. R. Meyer, and K. R. Stapelfeldt (2023). “Large Interferometer For Exoplanets (LIFE): X. Detectability of currently known exoplanets and synergies with future IR/O/UV reflected-starlight imaging missions.” *A&A*, 678: A96. ISSN: 1432-0746. DOI: [10.1051/0004-6361/202347027](https://doi.org/10.1051/0004-6361/202347027). URL: <http://dx.doi.org/10.1051/0004-6361/202347027>

8. Matsuo, T., **F. Dannert**, R. Laugier, S. P. Quanz, and A. B. Kovačević (2023). “Large Interferometer For Exoplanets (LIFE): XI. Phase-space synthesis decomposition for planet detection and characterization.” *A&A*, 678: A97. ISSN: 1432-0746. DOI: [10.1051/0004-6361/202345927](https://doi.org/10.1051/0004-6361/202345927). URL: <http://dx.doi.org/10.1051/0004-6361/202345927>

9. Angerhausen, D., D. Pidhorodetska, M. Leung, J. Hansen, E. Alei, **F. Dannert**, J. Kammerer, S. P. Quanz, and E. W. Schwieterman (2024). “Large Interferometer For Exoplanets (LIFE). XII. The Detectability of Capstone Biosignatures in the Mid-infrared—Sniffing Exoplanetary Laughing Gas and Methylated Halogens.” *AJ*, 167 (3): 128. ISSN: 1538-3881. DOI: [10.3847/1538-3881/ad1f4b](https://doi.org/10.3847/1538-3881/ad1f4b). URL: <http://dx.doi.org/10.3847/1538-3881/ad1f4b>

10. Cesario, L., T. Lichtenberg, E. Alei, Ó. Carrión-González, **F. A. Dannert**, D. Defrère, S. Ertel, A. Fortier, A. García Muñoz, A. M. Glauser, J. T. Hansen, R. Helled, et al. (2024). “Large Interferometer For Exoplanets (LIFE): XIV. Finding terrestrial protoplanets in the galactic neighborhood.” *A&A*, 692: A172. ISSN: 1432-0746. DOI: [10.1051/0004-6361/202450764](https://doi.org/10.1051/0004-6361/202450764). URL: <http://dx.doi.org/10.1051/0004-6361/202450764>

11. Hansen, J., D. Angerhausen, S. P. Quanz, D. Vance, B. S. Konrad, E. O. Garvin, E. Alei, J. Kammerer, and **F. A. Dannert** (2025). “Detecting Atmospheric CO₂ Trends as Population-Level Signatures for Long-Term Stable Water Oceans and Biotic Activity on Temperate Terrestrial Exoplanets.” *arXiv*: [2505.23230](https://arxiv.org/abs/2505.23230)

12. Huber, P. A., **F. A. Dannert**, R. Laugier, T. Matsuo, A. M. Glauser, and S. P. Quanz (2025). “Robust Data Interpretation for Perturbed Nulling Interferometers via Proper Handling of Correlated Errors.” in submission to ApJ

Additionally, for my work on LIFE I contributed to the following conference proceeding papers:

13. Laugier, R., D. Defrère, A. Matter, S. Gross, B. Courtney-Barrer, **F. Dannert**, J. Woillez, O. Absil, and C. Dandumont (2022). “The expected performance of nulling at the VLTI down to 5 mas.” In: *Optical and Infrared Interferometry and Imaging VIII*. edited by A. Mérand, S. Sallum, and J. Sanchez-Bermudez. SPIE, page 76. DOI: [10.1117/12.2629292](https://doi.org/10.1117/12.2629292). URL: <http://dx.doi.org/10.1117/12.2629292>
14. Huber, P. A., **F. A. Dannert**, R. Laugier, T. Birbacher, A. M. Glauser, J. T. Hansen, and S. P. Quanz (2024). “Analytical and numerical instrumental noise simulations for the Large Interferometer For Exoplanets (LIFE).” in: *Optical and Infrared Interferometry and Imaging IX*. edited by S. Sallum, J. Sanchez-Bermudez, and J. Kammerer. SPIE, page 50. DOI: [10.1117/12.3018644](https://doi.org/10.1117/12.3018644). URL: <http://dx.doi.org/10.1117/12.3018644>
15. Glauser, A. M., S. P. Quanz, J. Hansen, **F. Dannert**, M. J. Ireland, H. Linz, O. Absil, E. Alei, D. Angerhausen, T. Birbacher, D. Defrère, A. Fortier, et al. (2024). “The Large Interferometer For Exoplanets (LIFE): a space mission for mid-infrared nulling interferometry.” In: *Optical and Infrared Interferometry and Imaging IX*. edited by S. Sallum, J. Sanchez-Bermudez, and J. Kammerer. SPIE, page 48. DOI: [10.1117/12.3019090](https://doi.org/10.1117/12.3019090). URL: <http://dx.doi.org/10.1117/12.3019090>

In the context of the ERIS commissioning, I contributed to two peer-reviewed publications and two conference proceeding papers:

16. Davies, R., O. Absil, G. Agapito, A. Agudo Berbel, A. Baruffolo, V. Biliotti, M. Black, M. Bonaglia, M. Bonse, R. Briguglio, P. Campana, Y. Cao, L. Carbonaro, A. Cortes, G. Cresci, Y. Dallilar, **F. Dannert**, et al. (2023). “The Enhanced Resolution Imager and Spectrograph for the VLT.” *A&A*, 674: A207. ISSN: 1432-0746. DOI: [10.1051/0004-6361/202346559](https://doi.org/10.1051/0004-6361/202346559). URL: <http://dx.doi.org/10.1051/0004-6361/202346559>

17. Kenworthy, M. A., **F. A. Dannert**, J. Hayoz, D. Doelman, B. J. Sutcliffe, P. Liu, F. Snik, M. Bonse, S. Quanz, C. Keller, and X. Chen (2025). “The ERIS gvAPP Coronagraph: theoretical and on-sky performance.” in preparation
18. Kravchenko, K., Y. Dallilar, O. Absil, A. A. Berbel, A. Baruffolo, M. Bonse, A. Buron, Y. Cao, A. Cortes, **F. Dannert**, R. Davies, R. J. De Rosa, et al. (2022). “First on-sky results of ERIS at VLT.” in: *Ground-based and Airborne Instrumentation for Astronomy IX*. edited by C. J. Evans, J. J. Bryant, and K. Motohara. SPIE, page 211. DOI: [10.1117/12.2629258](https://doi.org/10.1117/12.2629258). URL: <http://dx.doi.org/10.1117/12.2629258>
19. Orban de Xivry, G., O. Absil, R. De Rosa, M. Bonse, **F. Dannert**, J. Hayoz, P. Grani, A. Puglisi, A. Baruffolo, B. Salasnich, R. Davies, A. M. Glauser, et al. (2024). “The VLT/ERIS vortex coronagraph: design, pointing control, and on-sky performance.” In: *Adaptive Optics Systems IX*. edited by D. Schmidt, E. Vernet, and K. J. Jackson. SPIE, page 42. DOI: [10.1117/12.3018265](https://doi.org/10.1117/12.3018265). URL: <http://dx.doi.org/10.1117/12.3018265>

My work on mid-infrared high contrast imaging observations lead to contributions to the following peer-reviewed publications:

20. Sartori, L. F., C. Lovis, J.-B. Delisle, M. Lendl, G. Cugno, A. Boehle, **F. Dannert**, A. Krenn, J. L. Gubler, and S. P. Quanz (2023). “L 363-38 b: A planet newly discovered with ESPRESSO orbiting a nearby M dwarf star.” *A&A*, 670: A42. ISSN: 1432-0746. DOI: [10.1051/0004-6361/202244347](https://doi.org/10.1051/0004-6361/202244347). URL: <http://dx.doi.org/10.1051/0004-6361/202244347>
21. Bonse, M. J., E. O. Garvin, T. D. Gebhard, **F. A. Dannert**, F. Cantalloube, G. Cugno, O. Absil, J. Hayoz, J. Milli, M. Kasper, and S. P. Quanz (2023). “Comparing Apples with Apples: Robust Detection Limits for Exoplanet High-contrast Imaging in the Presence of Non-Gaussian Noise.” *AJ*, 166 (2): 71. ISSN: 1538-3881. DOI: [10.3847/1538-3881/acc93c](https://doi.org/10.3847/1538-3881/acc93c). URL: <http://dx.doi.org/10.3847/1538-3881/acc93c>
22. Hayoz, J., G. Cugno, S. P. Quanz, P. Patapis, E. Alei, M. J. Bonse, **F. A. Dannert**, E. O. Garvin, T. D. Gebhard, B. S. Konrad, and L. F. Sartori (2023). “CROCODILE: Incorporating medium-resolution spectroscopy of close-in directly imaged exoplanets into atmospheric retrievals via cross-correlation.” *A&A*, 678: A178. ISSN: 1432-0746. DOI: [10.1051/0004-6361/202245752](https://doi.org/10.1051/0004-6361/202245752). URL: <http://dx.doi.org/10.1051/0004-6361/202245752>

23. Cugno, G., P. Patapis, A. Banzatti, M. Meyer, **F. A. Dannert**, T. Stolker, R. J. MacDonald, and K. M. Pontoppidan (2024). “Mid-infrared Spectrum of the Disk around the Forming Companion GQ Lup B Revealed by JWST/MIRI.” *ApJL*, 966 (1): L21. ISSN: 2041-8213. DOI: [10.3847/2041-8213/ad3cbc](https://doi.org/10.3847/2041-8213/ad3cbc). URL: <http://dx.doi.org/10.3847/2041-8213/ad3cbc>
24. Hayoz, J., M. J. Bonse, **F. Dannert**, E. O. Garvin, G. Cugno, P. Patapis, T. D. Gebhard, W. O. Balmer, R. J. De Rosa, A. Agudo Berbel, Y. Cao, G. Orban de Xivry, et al. (2025). “High-contrast spectroscopy with the new VLT/ERIS instrument: Molecular maps and radial velocity of the gas giant AF Lep b.” *A&A*, 698: A87. ISSN: 1432-0746. DOI: [10.1051/0004-6361/202453297](https://doi.org/10.1051/0004-6361/202453297). URL: <http://dx.doi.org/10.1051/0004-6361/202453297>
25. Bonse, M. J., T. D. Gebhard, **F. A. Dannert**, O. Absil, F. Cantalloube, V. Christiaens, G. Cugno, E. O. Garvin, J. Hayoz, M. Kasper, E. Matthews, B. Schölkopf, et al. (2025). “Use the 4S (Signal-Safe Speckle Subtraction): Explainable Machine Learning Reveals the Giant Exoplanet AF Lep b in High-contrast Imaging Data from 2011.” *AJ*, 169 (4): 194. ISSN: 1538-3881. DOI: [10.3847/1538-3881/adab79](https://doi.org/10.3847/1538-3881/adab79). URL: <http://dx.doi.org/10.3847/1538-3881/adab79>